



Wafir — Technical Report (AI Readiness Hackathon – KSA)

Team name: wafir team · **Track:** Finance (household energy budgeting / FinTech)

Solution name: Wafir — Proactive Bill Prediction and Budget Planning

Contact / Org: King Faisal University — Budget Planning · **Designation:** Students

Team Members

- Mohammed Nadher Aboalrejal — Mentor — aboalrejal.ai@gmail.com — Technical Lead, System Architect, Technical Report Author
- Fatima Alsultan (KFU, Chemical Engineering, 2nd Year) — fatima.alsultan2105@gmail.com — Document research, concept ideation, policy gap analysis, report structuring
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- Jorry Alfalah (KFU, Electrical Engineering, 2nd Year) — Jurryraed90@hotmail.com — App UI design, logo design
- Noor Alshammari (KFU, Chemical Engineering, 2nd Year) — noornaser.sh1@gmail.com — App UI templates, logo design

Cap: ≤5 pages. Demo telemetry is **synthetic** (hackathon use only); Knowledge Base documents are authentic public sources. This report describes only capabilities present in the repository (demo path + optional Supabase). Limitations are stated explicitly.

Project resources

- Live Demo: <https://wafir.aboalrejal.com/>
- GitHub: <https://github.com/aboalrejal-ai/wafir>
- Demo video: <https://www.youtube.com/watch?v=bRYwjbx9t4>

1. Introduction

Households in Saudi Arabia often discover electricity overspend only when the monthly bill arrives. Peak summer cooling makes this worse. **Wafir** is an Arabic RTL personal FinTech app that forecasts end-of-month SAR spend from **simulated** meter-like kWh plus weather context, compares the forecast to a user budget, and raises graduated alerts.

The pipeline follows **ITU-T Y.3172** (SRC → C → PP → M → P → D → SINK, plus Sandbox). Policy grounding uses a verified Knowledge Base of Saudi and ITU public sources (knowledge-base.json, 42 VERIFIED records + 1 ISO benchmark), compiled from three independent deep-research passes in docs/research/. Framework names follow ITU AI Readiness Report 2.0 (kb/framework/dimensions.json).

Honest limit: meter ingest is simulated; the forecast is rule-based seasonal projection, not a trained neural model.

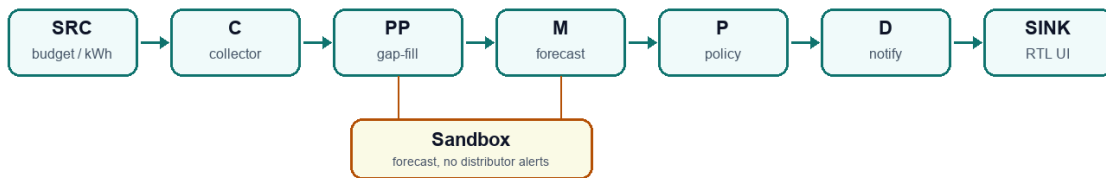
2. Use case and gaps in existing solutions

Problem. Families lack proactive, explainable energy-budget control tied to local tariffs and heatwaves.
Beneficiaries: Saudi households managing a monthly electricity budget; secondary value for energy/privacy policy readers via the gap matrix.

Gap in existing tools	How Wafir responds (implemented)
Late bill shock	Seasonal forecast vs budget; Level-2 warning if predicted SAR > budget or temperature ≥40°C
Opaque advice	RAG over verified chunks with citation URLs; if nothing matches: Arabic INSUFFICIENT_EVIDENCE, no invented law
No privacy gate	PDPL consent screen before processing; preprocessor pseudonymizes household id before ML export
Ads from consumption	Deterministic policy node blocks targeted_ads_from_consumption (PDPL-ADS-001)
No orchestration evidence	Demo path SRC→C→PP→M→P→D with audit log; reproducible via pnpm demo

3. Y.3172 architecture (mapped documents)

Sandbox is a Y.3172 supporting component (parallel validation). MLFO is seasonal-profile switching **inside M**, not an eighth official node. Y.3172 ↔ Readiness (ITU summary): D5 workflows, D10 policies, D13 infrastructure.



ITU-T Y.3172 · SRC → C → PP → M → P → D → SINK · Sandbox parallel to M

Node	Role in Wafir	Supporting document
SRC	Budget 500 SAR, simulated kWh, weather (OpenWeather if key set, else simulated)	SERA residential tariff
C	Demo collector / heatwave aggregation (simulateMeterReading). No consumer AMI API — GAP-01	SEC data sharing
PP	Gap-fill missing daily kWh; hash household id (pseudonymization, not full anonymization)	PDPL destruction / pseudonymisation
M + MLFO	Rule-based SAR forecast; summer profile at ≥38°C (method reference, not trained weights)	MDPI KSA energy forecasting
Sandbox	isSandbox: forecast without distributor alerts	ITU-T Y.3172
P	Budget L1/L1b/L2 + ads guard (no LLM in the decision)	PDPL official text + PDPL-ADS-001
D	In-app + Web/Local notifications (FCM needs keys)	SDAIA AI Ethics (human oversight)
SINK	Arabic RTL React UI. SAMA analog; not Open Banking integration	SAMA consumer protection

4. ITU AI Readiness 2.0 — six factors and selected dimensions

Official names: kb/framework/dimensions.json (ITU AI Readiness Report 2.0). PDPL / purpose limitation maps to **D10**, not D8.

Factor	Wafir evidence
Data	Public Saudi/ITU documents in knowledge-base.json (not private household dumps)
Research	Three deep-research passes in docs/research/ (ChatGPT, Gemini, Perplexity), then human merge
Deployment Support	Vite SPA on Hostinger; optional Supabase; Capacitor wrapper
Standards	Y.3172 node mapping + ITU-T Y.3172 citation
Open Source and Code	Public GitHub https://github.com/aboalrejal-ai/wafier
Sandbox Environments	isSandbox heatwave path — forecast without pushing alerts

ID	Official ITU dimension	Wafir evidence
D5	Level of Integration of AI in Workflows	Pipeline audit, graduated alerts, pnpm demo
D6	Human Interface	Arabic RTL UI + AI assistant with citations
D7	Strategy Alignment	KB NSDAI-001 (Vision 2030 / SDAIA NSDAI PDF). Use-case aligned to national strategy; we do not claim to implement NSDAI
D8	Collaboration with AI	HITL: 2-hour alert snooze on Profile; user question shapes RAG
D10	AI & Policies	Six verdicts; SC-03 policy sandbox; GAP-01..06; PDPL consent + purpose limitation
D11	AI for Inclusion	Arabic-first interface; local SERA/KAPSARC context
D13	Digital Infrastructure	Hosted SPA; optional Supabase; simulated meter as Y.3172 SRC stand-in

5. Three evaluation scenarios + Policy Gap Matrix

Reproducible proof: pnpm demo (or node scripts/run-demo.mjs). Demo data are synthetic.

SC-01 — Compliant RAG (scenarios/sc-01-compliant-rag.json)

Scenario. User asks in the AI Assistant: كيف أوفر في فاتورة الكهرباء؟ **Step 1.** The household submits a normal saving question. Context is synthetic. **Step 2.** Wafir retrieves verified KB chunks by keyword matching over curated records (not a live web search). If no chunk matches, the assistant returns لا توجد أدلة كافية... (INSUFFICIENT_EVIDENCE) and

invents no URLs. **Step 3.** For this query the grounded reply gives qualitative saving tips (runtime, efficient AC / SASO 2663, insulation, LED, night-time loads) with citation URLs. Expected verdict: **COMPLIANT**. Tips are not a guaranteed SAR saving.

SC-02 — Operational failure (scenarios/sc-02-pp-gap-fill.json)

Scenario. In a 7-day kWh series, **day 4 is missing** (kwh: null); other days are 29.67. **Step 1.** Collector/preprocessor sees an incomplete series (simulated meter gap, not a live AMI outage). **Step 2.** The missing day is filled from neighbouring values (demo expected fill = 29.67). Household id is pseudonymized for export. **Step 3.** Forecast and dashboard continue; the audit log records the gap-fill; the app does not crash. Expected: graceful degradation. The filled value is not claimed as a real meter reading.

SC-03 — Controversy / targeted ads (scenarios/sc-03-ads-controversy.json)

Scenario. A dummy provider requests targeted_ads_from_consumption. **Step 1.** The request arrives as a policy check (About → controversy scenario), not as a live ad SDK. **Step 2.** Policy node evaluateKbGuardPolicy runs **deterministically** (no LLM). **Step 3.** Result **VIOLATION / BLOCK_DATA_USE / PDPL-ADS-001**. In-app notice: منع الإعلانات — حارس سياسة KB. Audit stores the verdict and the PDPL knowledge-center URL. **Step 4 (corrective).** Consumption data is not released for ads. This is a **Wafir purpose-limitation control**. PDPL Art.26 allows consented marketing of non-sensitive data; we do not claim a statutory total advertising ban. HITL snooze remains available on Profile for budget alerts.

Classifications below are from sources **we reviewed**; they are not a claim that Saudi law is silent in every forum.

ID	Type	Gap	Recommendation
GAP-01	potential_gap	No consumer-authorized AMI API	Open-Utility style authorization (like Open Banking)
GAP-02	potential_gap	AI forecast liability allocation	Sector guidance; general civil law may apply (CIVIL-LAW-120)
GAP-03	potential_gap	Energy-sector AI data rules	PDPL + SERA PDP under explicit sector policy
GAP-04	potential_gap	No mandatory energy algorithm-audit standard	Y.3172 sandbox + ISO 42001 as a benchmark (not a certification)
GAP-05	potential_gap	No machine-readable law repository	Government structured-regulation API
GAP-06	ambiguity	Consumption → ads repurposing	Explicit PDPL/sector guidance; Wafir blocks silent reuse

- GAP-01.** In the sources we reviewed, we did not find a consumer-authorized AMI API comparable to Open Banking that would let Wafir pull live meter data with user consent. Wafir therefore uses simulated kWh. Recommendation: an Open-Utility style authorization framework.
- GAP-02.** We did not find sector-specific rules allocating liability when an energy forecast is wrong. General civil-law harmful-act principles may still apply (CIVIL-LAW-120). Recommendation: sector guidance; until then Wafir labels forecasts as estimates.
- GAP-03.** PDPL and SERA data-protection pages exist, but energy-AI processing (forecasting, sharing, secondary use) is not spelled out as a sector playbook in the sources we checked. Recommendation: PDPL + SERA PDP under explicit sector policy.
- GAP-04.** We did not find a mandatory energy-sector algorithm-audit standard for household bill models. Recommendation: use the Y.3172 sandbox pattern and ISO/IEC 42001 as a benchmark, not a claimed certification.
- GAP-05.** Official texts are published as HTML/PDF, not as a queryable regulation API. Wafir therefore curates verified chunks rather than scraping live law. Recommendation: a government structured-regulation API.
- GAP-06.** PDPL purpose limitation and Art.25–27 constrain marketing, but Art.26 still allows consented marketing of non-sensitive data. Whether electricity-consumption profiles may be reused for ads is not spelled out for this use case. Wafir blocks silent repurposing by policy (SC-03). Recommendation: explicit PDPL/sector guidance on energy-data repurposing.

6. Knowledge base (authentic sources) and repository links

Full corpus: knowledge-base.json (42 VERIFIED + 1 ISO benchmark). Ten primary citations for judges:

#	Document	Authority	URL
1	PDPL official EN (Art.4–6, 10–15, 25–27)	SDAIA	sdaia.gov.sa PDF
2	PDPL Art.26 direct marketing	SDAIA	dgp.sdaia.gov.sa PDPL2
3	AI Ethics Principles v1.0	SDAIA	ai-principles-EN.pdf
4	NSDAI (Vision 2030)	SDAIA	NSDAI.pdf
5	SERA residential consumption tariff	SERA	sera.gov.sa tariff
6	SEC data sharing	SEC	se.com.sa Open Data
7	KBEAT — Building Energy Assessment Tool	KAPSARC	apps.kapsarc.org/kbeat

#	Document	Authority	URL
8	Financial Consumer Protection Principles (analog)	SAMA	rulebook.sama.gov.sa
9	ITU-T Y.3172 ML pipeline architecture	ITU	itu.int Y.3172
10	ISO/IEC 42001:2023 (benchmark only)	ISO	iso.org 81230

Deliverable	Location
GitHub	https://github.com/aboalrejal-ai/wafier
Live demo	https://wafier.aboalrejal.com/
Demo video	https://www.youtube.com/watch?v=bRYwjbx9t4
Knowledge base	knowledge-base.json
Scenarios	scenarios/
Reproducible demo	pnpm demo

Distributor: in-app + Web Notification API + Capacitor Local Notifications.